SpecSwin3D: Generating Hyperspectral Imagery From Multispectral Data via Transformer Networks and Curriculum-Based Cascade Training

Tang Sui, Songxi Yang, and Qunying Huang

***Abstract*—Multispectral (MS) and hyperspectral (HS) imageries are widely used in agriculture, environmental monitoring, and urban planning due to their complementary spatial and spectral characteristics. A fundamental tradeoff persists: MS imagery offers high spatial but limited spectral resolution, while HS imagery provides rich spectra at lower spatial resolution. Prior HS generation approaches often struggle to jointly preserve spatial detail and spectral fidelity. In response, we propose *SpecSwin3D*, a Swin Transformer-based model that generates HS imagery from MS inputs while preserving both spatial and spectral qualities. Specifically, SpecSwin3D uses five MS input bands to generate 224 HS bands at the same spatial resolution. In addition, existing methods that construct all HS bands using a single global model suffer from increasing generation errors for bands that are spectrally distant from the input bands, while training separate models (band-specific) for individual bands is computationally intensive. To address this tradeoff, we propose a curriculum-based cascade training strategy that progressively expands the spectral range from easier, MS-adjacent bands to more challenging, spectrally distant bands. This approach enables stable learning from spectrally proximal to distal bands and improves reconstruction fidelity for each individual band while significantly improving computational efficiency. Moreover, we design an optimized band sequence that strategically repeats and orders the five selected MS bands to better capture pairwise relations of the spectrum within a 3-D shifted-window Transformer framework. Quantitatively, our model achieves a peak signal-to-noise ratio (PSNR) of 35.84 dB, spectral angle mapper (SAM) of 2.39°, and structural similarity index metric (SSIM) of 0.96, outperforming state-of-the-art deep learning based approach by +5.8 dB in PSNR and reducing ERGAS by more than half. Beyond HS band generation, we further demonstrate the practical value of SpecSwin3D on two downstream tasks, including land-use classification and burned area segmentation, achieving satisfactory results with enhanced both spatial and spectral resolutions. Although SpecSwin3D uses a single unified model, cascade training substantially reduces the training budget for the majority of target bands (e.g., from 80 to 20 epochs at later levels), resulting in about 75% lower training cost compared with uniform training.**

***Index Terms*—Cascade training, curriculum learning, deep learning, hyperspectral (HS) image generation, HS imagery reconstruction, remote sensing, Swin Transformer, SwinUNETR, Transformer.**

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I. Introduction

MULTISPECTRAL (MS) and hyperspectral (HS) imagery play a critical role in remote sensing applications such as precision agriculture [1], change detection [2], and land-use classification [3], due to their ability to capture complementary spatial and spectral information [4]. MS imagery typically contains a limited number of discrete bands (e.g., blue, green, red, and near-infrared (NIR) within 400–900 nm) with high spatial resolution, whereas HS imagery provides hundreds of contiguous spectral bands over a broader wavelength range (e.g., 400–2500 nm) at lower spatial resolution [5]. This fine-grained spectral representation enables detailed discrimination of materials and biochemical properties, supporting diverse applications including agriculture [6], [7], cultural heritage analysis [8], food quality assessment [9], [10], and military surveillance [11], [12]. To mitigate the inherent tradeoff between spatial and spectral resolutions, various HS imagery generation (HIG) techniques, such as pan-sharpening [13], matrix factorization (MF) [14], [15], [16], and recently convolutional neural networks (CNNs) [17], [18], [19], have been developed to reconstruct HS imagery from MS observations. Early efforts focused on pan-sharpening and statistical fusion methods, employing intensity matching and linear transforms such as IHS, PCA, and Gram–Schmidt orthogonalization [13], [20], [21], [22], laying a solid foundation for subsequent HS–MS fusion methods that explicitly model spectral relationships [23], [24].

MF-based approaches further advanced HIG by decomposing HS data into spectral bases and coefficients, using sparse representation or low-rank assumptions to capture spectral redundancy [14], [15], [16]. Although effective, MF methods often rely on carefully tuned observation models and incur high computational cost, limiting scalability [25], [26], [27], [28], [29]. More recently, deep learning has become the dominant paradigm for HIG. CNN-based models leverage local spatial context for spectral reconstruction [17], [18], [19], [30], [31], [32]. These methods can be categorized into one- and two-branch approaches. One-branch methods (e.g., PanNet [31]) combine features from MS imagery and HS imagery and use a single CNN branch to predict high spatial resolution HS imagery, whereas two-branch CNN-based approaches [33] use separate networks for spatial and spectral feature extraction before fusion. Transformer-based methods further improve performance by capturing long-range spectral–spatial dependencies, with representative works

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including Fusformer [34], MSST-Net [35], and subsequent efficient attention variants [36], [37], [38]. In parallel, generative models such as GANs and diffusion models have also been explored to enhance spectral realism, though often at the cost of training instability or high inference complexity [39], [40], [41]. Despite these advances, existing HIG methods struggle to simultaneously preserve spatial detail and spectral fidelity. Pan-sharpening methods are prone to spectral distortion [42], MF-based approaches are sensitive to model assumptions, and CNN-based models are limited in capturing long-range spectral dependencies. Although Transformer-based methods improve global modeling, most operate on 2-D token windows and do not explicitly exploit the intrinsic 3-D spectral–spatial structure of HS data. Moreover, our preliminary investigation using both CNN- and Transformer-based HIG models [34], [35], [43], [44], [45] reveals that reconstructing all HS bands with a single global model leads to progressively larger errors for bands spectrally distant from the MS inputs. As illustrated in Fig. 1, bands proximal to the MS wavelengths are reconstructed accurately, whereas distant bands exhibit pronounced deviations due to weak supervision and suboptimal gradient allocation under a global loss formulation. While band-specific models could alleviate this issue, they are computationally infeasible in practice. To address these challenges, we propose SpecSwin3D, a Transformer-based HS generator built upon a 3-D Swin Transformer architecture [46] that jointly models spectral–spatial dependencies across channel and spectral-depth dimensions. Furthermore, a curriculum-based cascade training strategy is introduced to progressively reconstruct HS bands according to physical spectral priors and statistical dependence with MS inputs, enabling stable extrapolation from sparse MS observations to the full HS spectrum. The model is trained and evaluated on a large-scale simulated HS–MS dataset pairs derived from the AVIRIS repository [47]. The main contributions of this work are summarized as follows.

1) We propose SpecSwin3D, a 3-D SwinUNETR-based Transformer for HIG that captures complex spectral characteristics and correlation between different bands from HS imagery and learn fine spatial details from HSR MS imagery, effectively integrating the complementary strengths of both imaging types.
2) An optimized band sequence strategy is proposed to maximize local interband interaction while ensuring full pairwise coverage in the spectral (depth) dimension, which enhances spectral feature extraction under a 3-D shifted-window framework.
3) Six curriculum-based cascade training strategies are introduced to progressively handle spectral bands with varying distances from the input MS bands, enabling more accurate HS band generation with reduced computational cost while preserving spectral fidelity and spatial detail.
4) Extensive experiments on AVIRIS demonstrate consistent gains over state-of-the-art HIG baselines.
5) We validate downstream benefits on land-use classification and burned-area detection, showing that the generated HS bands provide substantial benefits in both spatial and spectral domains.

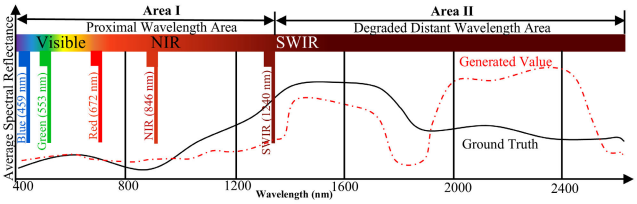


Fig. 1. Preliminary spectral reconstruction analysis with the red curve representing the generated spectrum and the black curve indicating the ground truth. Area I corresponds to wavelengths proximal to the MS bands with accurate reconstruction, whereas Area II highlights increased deviations for spectrally distant bands.

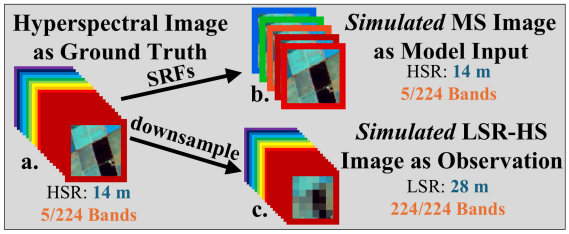


Fig. 2. Simulation of (b) HSR MS and (c) LSR HS bands using (a) HSR HS imagery. Five HS bands were selected to match the wavelengths of MS bands, while all HS bands were downsampled to simulate LSR HS bands.

II. Methods

A. Data Preprocessing

To ensure comparability across platforms, MS bands can generally be synthesized from HS imagery using spectral response functions (SRFs) (Fig. 2), where each MS band is modeled as a weighted integration of the HS spectrum and the weights can be estimated via a least-squares formulation [48]. In our setting, since both MS and HS data are derived from AVIRIS, we simulate the MS inputs by directly selecting five HS bands whose center wavelengths are closest to the desired MS bands (an extreme case of SRF-based synthesis, where each SRF degenerates to a delta function). We further downsample the original 14-m HS imagery to 28 m using bilinear interpolation to simulate the common low-spatial-resolution HS acquisition setting; this LSR HS imagery is used only for simulation and visualization and is not used as an input during model training. Specifically, we selected band 9 (459 nm, blue), band 20 (553 nm, green), band 30 (672 nm, red), band 40 (846 nm, NIR), and band 52 (1240 nm, SWIR) as the simulated MS inputs. During training, each image was cropped into 128×128 pixel tiles model inputs. To improve model generalization, data augmentation was applied, including random rotations (-15° to 15°), random cropping (70%–90% of the original image size), and horizontal and vertical flipping. The dataset was stratified into training (70%), validation (15%), and test (15%) sets to preserve class distributions. The original HSR HS images at 14 m were treated as pseudo ground truth, while MS inputs were simulated using SRFs. The model was trained to learn the spectral mapping from the simulated MS inputs to the corresponding

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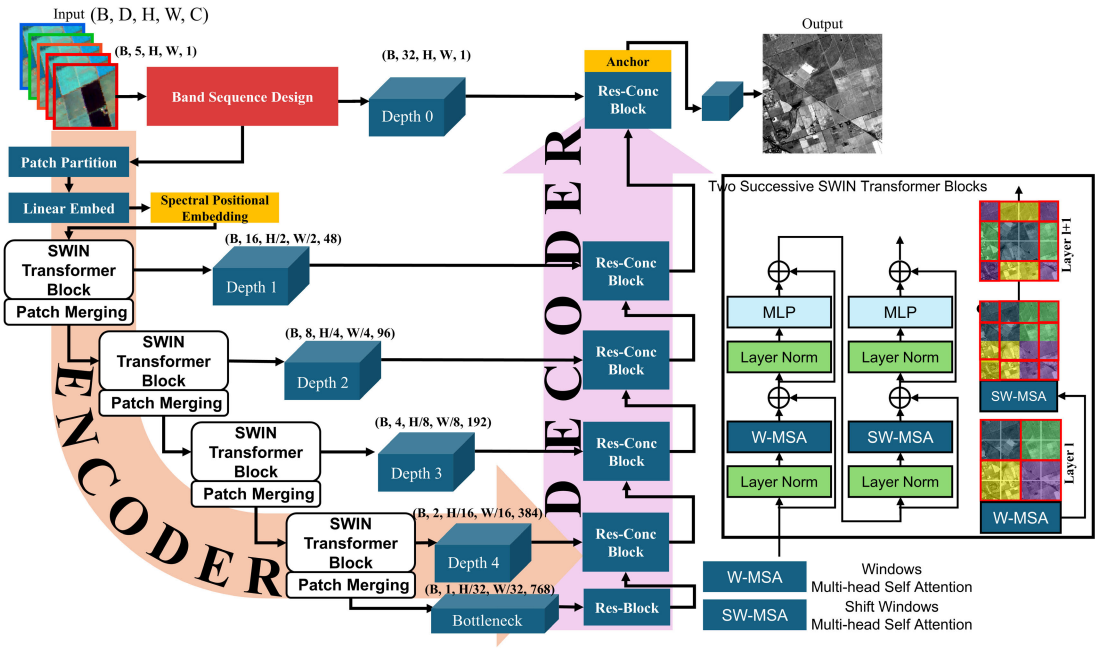


Fig. 3. Overview of the SpecSwin3D architecture: The input consists of MS images with five bands, which are first divided into nonoverlapping patches and embedded for efficient 3-D self-attention. Encoded features are progressively reconstructed through a multiscale hierarchical decoder with skip connections across different resolutions. The final output can be either a single band or multiple bands, depending on the reconstruction task.

HSR HS targets. After training, the model was evaluated by generating HS images from the simulated MS inputs, which were compared against the original HSR HS ground truth.

B. Model Architecture

SpecSwin3D is a 3-D Transformer-based encoder–decoder framework for HIG that models both local and long-range dependencies by capturing correlations within and across spectral slices, inspired by advances in volumetric imaging architectures [49], [50]. It builds on the SwinUNETR architecture [51], originally developed for 3-D brain tumor segmentation using magnetic resonance imaging (MRI), which processes volumetric MRI data composed of multiple slices to generate a tumor segmentation mask. The model effectively captures both spatial and interslice dependencies by modeling not only the correlations within individual slices but also the relationships across adjacent slices. The Swin Transformer architecture effectively leverages information across neighboring slices, enhancing segmentation performance [46].

The core idea of *SpecSwin3D* architecture is to reproject spectral bands into the depth dimension of a 3-D volume, treating neighboring wavelengths as adjacent slices. Analogous to volumetric imaging, where adjacent slices exhibit strong correlations, HS bands demonstrate high continuity across nearby wavelengths [52]. Exploiting this property with 3-D shifted-window attention, *SpecSwin3D* enhances reconstruction fidelity, especially for spectrally adjacent bands, while preserving fine spatial detail. Moreover, our 3-D volume is constructed through a deliberately designed sequence that accounts for interslice dependencies rather than relying on simple sequential band stacking (see Section II-C for details).

1) Swin3D Encoder: SpecSwin3D integrates a Swin Transformer encoder [46] within a U-Net framework to effectively capture hierarchical information from the input data. As depicted in Fig. 3, the model's input is first processed through a patch partitioning and linear embedding block, which transforms the image into 3-D tokens. In the encoding stage, two successive Swin Transformer blocks are employed to efficiently compute self-attention with reduced computational complexity. The window multihead self-attention (W-MSA) blocks compute self-attention within nonoverlapping windows. To enhance information exchange between windows, the shifted W-MSA (SW-MSA) mechanism introduces a 3-D cyclic-shifting window to facilitate information flow across adjacent windows and capture interchannel relationships. Following the Swin Transformer blocks, patch merging layers reduce the spatial dimensions (height, width, and depth) while increasing the channel (dimension) through linear transformations. This downsampling mechanism not only decreases spatial resolution but also enhances the model's capacity to capture long-range dependencies by utilizing more channels. This encoding phase significantly expands the global receptive field efficiently.

2) Decoder and Output Strategy: The decoder adopts a U-Net-style architecture to progressively upsample multiscale features and reconstruct the HS cube at full spatial resolution. Residual upsampling blocks and skip connections fuse local spatial details with global contextual information across scales.

Training is guided by a composite spectral supervision objective that jointly enforces reconstruction accuracy, spectral shape consistency, physical identifiability, and spectral smoothness, enabling physically plausible and spectrally coherent HIG across the entire wavelength range. The detailed loss formulation is provided in Section II-E.

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C. Band Sequence Design for 3-D Shifted Window

To effectively model interband relationships within the MS inputs (*bands* [9, 20, 30, 40, 52], as defined in Section II-A), we designed a 3-D input of shape $H \times W \times D \times S$, where H and W denote the spatial dimensions, D is the depth (i.e., the number of spectral slices), and S is the number of channels per slice. In our implementation, the depth dimension is used to organize the sequence of spectral bands, allowing the SpecSwin3D's SW-MSA mechanism to capture 3-D context across adjacent spectral slices. The overall 3-D token organization and shifted-window processing pipeline are illustrated in Fig. 4. To ensure that all $\binom{5}{2} = 10$ unique band pairs among the five selected input bands co-occur within at least one adjacent slice pair in the depth dimension, we designed a compact input sequence of length $D = 16$: [9, 20, 30, 40, 52, 20, 40, 9, 30, 52, 9, 20, 30, 40, 52, 30]. This sequence ensures complete pairwise coverage.

D. Spectral Positional Embedding

In addition to the optimized band sequence, we incorporate a *spectral positional embedding (SPE)* to encode absolute wavelength information for each MS band. While the handcrafted ordering preserves the local spectral continuity needed for 3-D shifted-window attention, it does not explicitly provide the model with the physical wavelength location of each band. SPE complements this by injecting wavelength-aware semantic information into the Transformer input tokens.

Given an MS band with central wavelength λ_i, we generate a learnable embedding vector $\mathbf{e}_i \in \mathbb{R}^d$, where d denotes the token embedding dimension of the SpecSwin3D encoder, using a lightweight two-layer MLP, i.e., $\mathbf{e}_i = \mathrm{MLP}(\lambda_i)$, where λ_i serves as the scalar spectral coordinate. The embedding is added to the input token features before entering the encoder as $\tilde{\mathbf{x}}_i = \mathbf{x}_i + \mathbf{e}_i$. Unlike classical sinusoidal positional encodings that assume uniform spatial intervals, SPE does not rely on any implicit spatial uniformity. Instead, it adapts to the nonuniform, physically meaningful distribution of wavelengths across the electromagnetic spectrum. In contrast to continuous wavelength encodings $f(\lambda)$ used in previous work, SPE is directly optimized together with the network parameters, enabling the model to learn nonlinear spectral relationships such as absorption features or reflectance changes that vary across wavelength ranges. Through this design, SPE provides a global spectral anchor for each MS band while maintaining compatibility with our optimized local band sequence: the former supplies absolute spectral location, and the latter ensures contiguous spectral neighborhoods for shifted-window attention, jointly improving the model's ability to capture both local and global spectral dependencies.

E. Spectral Regularization

To mitigate spectral hallucination during the HIG process, SpecSwin3D introduces four complementary spectral constraints that enforce physical identifiability, manifold consistency, and local spectral continuity.

1) Anchor Consistency Constraint: Because each MS band is a weighted integration of HS reflectance shaped by the sensor's SRF, the predicted HS spectrum $\hat{\mathbf{S}}_{\mathrm{HS}}$ should reproduce the original MS observation [53]. This is enforced by $\hat{\mathbf{S}}_{\mathrm{MS}} = \mathbf{W}_a \hat{\mathbf{S}}_{\mathrm{HS}}$, where $\mathbf{W}_a \in \mathbb{R}^{5\times 224}$ is the SRF-derived projection matrix. The corresponding anchor consistency loss is defined as $\mathcal{L}_{\mathrm{anchor}} = \|\hat{\mathbf{S}}_{\mathrm{MS}} - \mathbf{S}_{\mathrm{MS}}\|_1$, which prevents physically implausible spectra.

2) PCA-Based Spectral Manifold Constraint: Natural HS signals lie on a low-dimensional manifold. We compute PCA on the training HS data and project both the prediction and ground truth onto the first K eigenvectors [54], [55], yielding $\hat{\mathbf{z}} = \mathbf{P}^\top \hat{\mathbf{S}}_{\mathrm{HS}}$ and $\mathbf{z} = \mathbf{P}^\top \mathbf{S}_{\mathrm{HS}}$. A manifold-alignment loss then constrains predictions to remain within the physically valid spectral manifold via $\mathcal{L}_{\mathrm{pca}} = \|\hat{\mathbf{z}} - \mathbf{z}\|_1$.

3) SAM Loss: To enforce spectral shape fidelity in a scale-invariant manner, we incorporate the classical spectral angle mapper (SAM) loss defined as $\mathcal{L}_{\mathrm{sam}} = \arccos((\langle \hat{\mathbf{S}}_{\mathrm{HS}}, \mathbf{S}_{\mathrm{HS}} \rangle)/(\|\hat{\mathbf{S}}_{\mathrm{HS}}\| \, \|\mathbf{S}_{\mathrm{HS}}\|))$.

4) Spectral Smoothness Loss: Real-world reflectance spectra exhibit strong local continuity. To encourage local spectral smoothness and discourage hallucinated spikes, we introduce a first-order spectral difference loss defined as $\mathcal{L}_{\mathrm{smooth}} = (1)/(B-1) \sum_{\lambda=1}^{B-1} \|\hat{\mathbf{S}}_{\mathrm{HS}}[:, \lambda+1] - \hat{\mathbf{S}}_{\mathrm{HS}}[:, \lambda]\|_1$.

Finally, the total training objective integrates all constraints as $\mathcal{L}_{\mathrm{total}} = \mathcal{L}_1 + \alpha \mathcal{L}_{\mathrm{sam}} + \beta \mathcal{L}_{\mathrm{smooth}} + \gamma \mathcal{L}_{\mathrm{anchor}} + \delta \mathcal{L}_{\mathrm{pca}}$.

F. Curriculum-Based Cascade Training Strategy and Fine-Tuning

Our objective is to generate 224 HSR HS bands from only five HSR MS inputs. A straightforward modeling choice is to use a single global network to jointly predict all HS bands, treating the output as a multichannel image. However, such a formulation optimizes an average loss over all bands, which often leads to suboptimal gradients for individual spectral channels, particularly for bands that are spectrally distant from the MS inputs (Fig. 1). As a result, reconstruction accuracy tends to degrade in these distant spectral regions. An alternative extreme is to train independent models for each HS band, which allows band-specific optimization but is computationally prohibitive in practice. These limitations motivate the proposed cascade training strategy, which balances per-band optimization and computational efficiency by progressively expanding the spectral prediction range within a single unified model.

1) Stage 1: Initial Training on the Easiest Band Set (Level 0): The first stage aims to establish a stable and robust spectral–spatial representation by training the model on the easiest HS bands. These bands are spectrally closest to the MS inputs and typically exhibit smoother spectral responses and weaker absorption effects, making them easier to reconstruct. Starting from such bands reduces optimization difficulty and provides a reliable initialization for subsequent spectral expansion.

To this end, we define a Level 0 band set that contains a small subset of HS bands with minimal spectral distance to the MS bands. During this stage, the model is trained only to predict these Level 0 bands while keeping the network

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TABLE I

SUMMARY OF CASCADE CURRICULUM STRATEGIES. EACH CELL LISTS THE TARGET-BAND CATEGORY AND THE CORRESPONDING BAND INDICES. BANDS ARE PROGRESSIVELY GROUPED FROM EASIER (LEVEL 0) TO MORE CHALLENGING (LEVEL 5) RECONSTRUCTION TARGETS

Strategy	Level 0	Level 1	Level 2	Level 3	Level 4	Level 5
Physical	RGB/NIR bands [0–8]	Transition bands [15,25,27]	SWIR sub-ranges [48,50,54,67,81,83,90]	Distant SWIR [108,125,135]	Far SWIR [155,162,175,185,189,210,218]	Remaining bands (all except Levels 0–4)
Spectral Physics	Absorption-free [70,71,72,80–84,103]	Weak absorption [73,74,75]	Moderate absorption [76–79,85,86,87]	Strong absorption [88,89,90]	Post-absorption [91–98]	Remaining bands (all except Levels 0–4)
Mutual Information	Highest MI [9,18,26–32]	High MI [16,19,25]	Moderate MI [8,10,14,15,17,20,24]	Low MI [11,12,13]	Very low MI [7,21,22,23,33,47,48]	Remaining bands (all except Levels 0–4)
Variance	Lowest variance [115,165-172]	Low variance [114,116,173]	Moderate variance [111,112,113,117,118,119,174]	High variance [164,175,176]	Highest variance [120,177,178,207,208,209,211]	Remaining bands (all except Levels 0–4)
Correlation	Highest correlation [16,19,20–25,35]	High correlation [15,17,18]	Moderate correlation [12,13,14,26–29]	Low correlation [11,30,31]	Very low correlation [8,9,10,32,33,34,36]	Remaining bands (all except Levels 0–4)
Uniform	Uniform spacing [0,7,15,23,31,38,46,54,62]	Uniform spacing [70,77,85]	Uniform spacing [93,101,109,116,124,132,140]	Uniform spacing [147,155,163]	Uniform spacing [171,179,186,194,202,210,218]	Remaining bands (all except Levels 0–4)

Note: Six cascade curriculum strategies are designed to rank HS bands from easier to more challenging reconstruction targets based on complementary physical and statistical criteria. Cascade Levels 0–4 contain 29 representative HS bands used for progressive curriculum training, while Level 5 includes the remaining HS bands that are introduced only during final fine-tuning.
Variance-based strategy: For each HS band b, spectral variability is quantified as $s_{\mathrm{var}}(b) = \mathrm{Var}(\mathbf{X}_{:,:,b})$, where $\mathbf{X}_{:,:,b}$ denotes the reflectance values of band b across all spatial pixels and training samples. Bands with higher variance typically exhibit richer spectral dynamics and stronger sensitivity to surface materials and illumination, and are therefore considered harder reconstruction targets.
Correlation-based strategy: Inter-sensor dependency is measured by the average Pearson correlation between each HS band b and the five MS input bands: $s_{\mathrm{corr}}(b) = \frac{1}{5}\sum_{i=1}^{5} \rho(\mathbf{X}_{:,:,b}, \mathbf{M}_{:,:,i})$, where $\mathbf{M}_{:,:,i}$ denotes the i-th MS band. Higher correlation indicates stronger linear predictability from MS inputs and lower reconstruction difficulty.
Mutual information-based strategy: To capture both linear and nonlinear dependencies, mutual information is computed as $s_{\mathrm{mi}}(b) = \frac{1}{5}\sum_{i=1}^{5} \mathrm{MI}(\mathbf{X}_{:,:,b}, \mathbf{M}_{:,:,i})$, estimated in a sample-wise manner over the training set. Bands with higher mutual information share more information content with MS inputs and are thus easier to reconstruct.
Physical strategy: HS bands are grouped according to coarse spectral regions (visible, NIR, SWIR) and ranked by spectral proximity to the MS wavelengths, reflecting physical continuity of reflectance spectra and sensor sensitivity.
Spectral-physics strategy: Bands are ranked using absorption-aware priors derived from known atmospheric and material absorption characteristics. Bands dominated by strong or unstable absorption features are treated as more challenging due to reduced signal-to-noise ratio and weaker MS–HS coupling.
Uniform strategy: Bands are selected with equal wavelength spacing as a reference curriculum without explicit physical or statistical priors, serving as a neutral baseline for comparison.

architecture identical to later stages. This initial training allows the model to learn fundamental spatial structures and local spectral correlations without being affected by the strong variability and ambiguity present in distant spectral regions. By restricting the training targets to the easiest bands in Stage 1, the model converges faster and achieves stable gradients, which is crucial for effective curriculum-based expansion in the subsequent stages.

2) Stage 2: Curriculum-Based Cascade Training on Representative Bands (Levels 1–4): After the model is stably initialized using the easiest spectral targets in Stage 1, we progressively expand the HIG task to more challenging HS bands through curriculum-based cascade training. This stage is designed to mitigate optimization instability and performance degradation that arise when spectrally distant bands are introduced too early. Rather than enforcing uniform group sizes across cascade levels, the number of bands introduced at each level is intentionally kept small and uneven. This design reflects the varying density and informativeness of spectral regions, and prioritizes introducing bands that remain spectrally or statistically connected to previously learned targets, ensuring smooth optimization and stable knowledge transfer [56], [57], [58]. Specifically, a set of representative HS bands is partitioned into cascade Levels 1–4 based on reconstruction difficulty, determined by spectral distance, statistical dependence with MS inputs, or physical absorption characteristics. As summarized in Table I, we explore six curriculum strategies, including uniform spacing, physical spectral grouping, spectral-physics priors, and three data-driven criteria based on variance, correlation, and mutual information. These strategies provide complementary perspectives for ranking HS bands from easier to more challenging reconstruction targets. During cascade training, the model architecture and parameters are shared across all levels. Starting from the pretrained Level 0 model, additional HS bands are progressively introduced as prediction targets from Level 1 to Level 4. At each stage, the model is initialized from the previous level, allowing the knowledge learned from the closer bands to be transferred to the more distant ones. This progressive expansion leverages the stronger spectral correlation among neighboring bands and promotes stable optimization as the target set grows. To balance convergence stability and computational efficiency, the number of training epochs decays by a factor of 0.9 at each successive level, resulting in 80, 72, 65, 58, and 53 epochs for Levels 0–4, respectively.

Fig. 5 illustrates the proposed cascade band grouping and training schedule along the spectral dimension. By gradually increasing spectral difficulty, the model learns long-range spectral dependencies in a controlled manner, reducing error accumulation for distant wavelengths and improving overall reconstruction robustness.

3) Stage 3: Fine-Tuning on Remaining HS Bands (Level 5): After curriculum-based cascade training on the representative HS bands (Levels 0–4), the model is further fine-tuned

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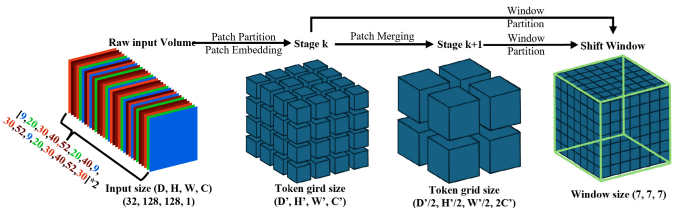


Fig. 4. Overview of shifted window, patch partition, embedding, and patch merging.

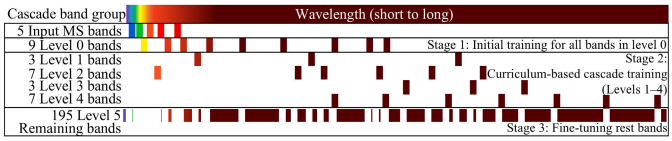


Fig. 5. Illustration of the proposed cascade band grouping and training schedule along the spectral dimension. Five MS bands are used as inputs. A subset of 29 HS target bands is progressively introduced from Level 0 to Level 4 based on spectral proximity and difficulty, followed by fine-tuning on the remaining 190 HS target bands (excluding MS inputs).

on the remaining HS bands (Level 5) to complete full-spectrum reconstruction. Rather than training these bands from scratch, fine-tuning is initialized from the pretrained parameters obtained in Stages 1 and 2. To balance computational efficiency and reconstruction accuracy, the fine-tuning budget for each band is adaptively determined based on its spectral similarity to the cascade-trained bands and its spectral distance. Specifically, a base number of fine-tuning epochs is first assigned according to the similarity score and then scaled by a spectral distance factor (SDF), defined as a monotonic function of the wavelength distance between the target band and the nearest cascade-trained band. The final number of fine-tuning epochs for band b is computed as $E_b = E_{\text{base}} \times \text{SDF}(b)$, where SDF($b$) increases with spectral distance, allocating more optimization steps to spectrally distant or weakly correlated bands. This adaptive fine-tuning strategy enables difficult spectral regions to receive sufficient learning capacity while preventing overfitting in bands that are already well constrained by the cascade training, resulting in stable and accurate reconstruction across the full HS range.

A clear contrast can be observed between Figs. 1 and 6. Without cascade training (Fig. 1), reconstruction accuracy deteriorates significantly for HS bands that are spectrally distant from the MS inputs. In contrast, when the proposed cascade training strategy is applied (Fig. 6), reconstruction performance on distant spectral bands is substantially improved and becomes more stable across the full spectral range. This comparison demonstrates that different HS bands exhibit inherently different reconstruction difficulties, and confirms the necessity of a structured cascade pyramid followed by targeted fine-tuning to achieve consistent and reliable HIG. The detailed cascade training discussion is provided in Section IV-B.

G. Evaluation Metrics

To evaluate the effectiveness of SpecSwin3D, we compared it against six established HIG deep learning methods: CASSIDiff [41] (task-adapted), MHF-Net [43], DBIN [44], Fusformer [34], MoG-DCN [45], and MSST-Net [35]. Five widely used metrics were employed to comprehensively evaluate the quality of the constructed HS bands: peak signal-to-noise ratio (PSNR) [59], ERGAS [60], [61], SAM [62], [63], Q Index [64], and structural similarity index metric (SSIM) [65]. Specifically, PSNR measures pixel-wise reconstruction fidelity by quantifying the logarithmic ratio between signal power and reconstruction error

$$\text{PSNR} = 10\log_{10}\left(\frac{\max(f)^2}{\text{MSE}}\right) \tag{1}$$

where

$$\text{MSE} = \frac{1}{\text{MNK}}\sum_{k=1}^{K}\sum_{i=1}^{M}\sum_{j=1}^{N}\left(f_k(i,j) - \hat{f}_k(i,j)\right)^2. \tag{2}$$

ERGAS evaluates global spectral–spatial consistency by normalizing reconstruction error with respect to band-wise mean intensity

$$\text{ERGAS} = 100 \cdot \left(\frac{1}{B}\sum_{b=1}^{B}\left(\frac{\text{RMSE}_b}{\mu_b}\right)^2\right)^{0.5}. \tag{3}$$

SAM assesses spectral shape similarity by computing the average spectral angle between reference and reconstructed spectra at each pixel:

$$\text{SAM} = \frac{1}{\text{MN}}\sum_{i=1}^{M}\sum_{j=1}^{N}\arccos\left(\frac{\langle f(i,j), \hat{f}(i,j)\rangle}{\|f(i,j)\|\,\|\hat{f}(i,j)\|}\right). \tag{4}$$

The Q index measures per-band similarity by jointly considering luminance, contrast, and correlation:

$$Q = \frac{1}{B}\sum_{b=1}^{B}\frac{(2\mu_b\hat{\mu}_b + C_1)(2\sigma_{b,b} + C_2)}{(\mu_b^2 + \hat{\mu}_b^2 + C_1)(\sigma_b^2 + \hat{\sigma}_b^2 + C_2)}. \tag{5}$$

SSIM evaluates structural similarity across spectral bands by comparing local statistics of the reference and reconstructed images

$$\text{SSIM} = \frac{1}{K}\sum_{k=1}^{K}\frac{(2\mu_k\hat{\mu}_k + C_1)(2\sigma_{k,k} + C_2)}{(\mu_k^2 + \hat{\mu}_k^2 + C_1)(\sigma_k^2 + \hat{\sigma}_k^2 + C_2)}. \tag{6}$$

Here, $f_k(i,j)$ and $\hat{f}_k(i,j)$ denote the reference and reconstructed pixel values at location (i,j) of band k; M, N, and K are the spatial dimensions and the number of HS bands; B denotes the total number of bands; μ, σ^2, and $\sigma_{(\cdot,\cdot)}$ represent the mean, variance, and covariance, respectively; and C_1 and C_2 are small constants for numerical stability.

III. Performance Evaluation

The proposed SpecSwin3D model is trained and evaluated using a comprehensive HS dataset from California and Nevada (Fig. 7), sourced from the publicly available AVIRIS repository [47]. The dataset consists of 3,000 orthorectified HS images with 14 m spatial resolution and 224 spectral bands spanning 400–2500 nm. We benchmarked our approach against six state-of-the-art HIG methods, including CASSIDiff [41], MHF-Net [43], DBIN [44], Fusformer [34], MoG-DCN

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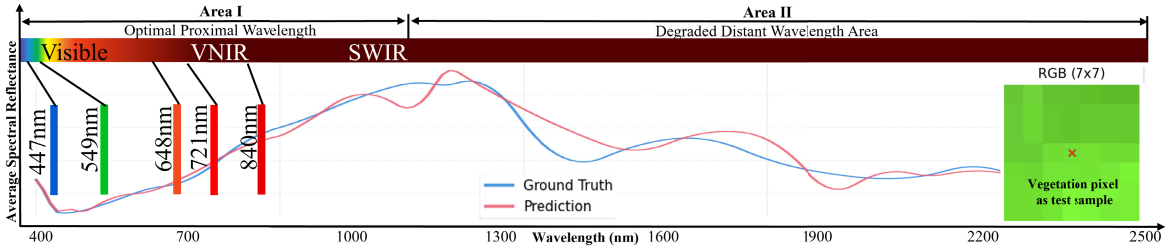


Fig. 6. Spectral reconstruction performance using physical strategy for a vegetation pixel, selected as the representative test case (inset: RGB patch). The orange curve denotes SpecSwin3D predictions, and the blue curve denotes the ground truth. Similar to Fig. 1, two wavelength regions are highlighted: Area I (proximal bands with high accuracy) and Area II (distant bands with reduced accuracy). Note that the y-axis is log-scaled to fit the full spectral range; this amplifies visual deviations, while absolute reconstruction errors remain small.

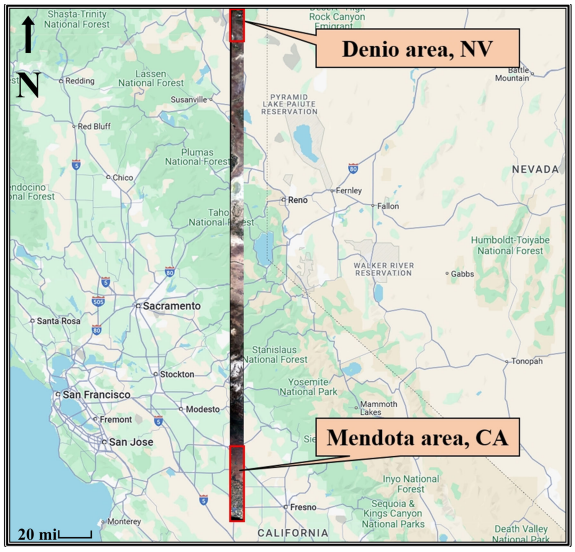


Fig. 7. Study area corresponding to the AVIRIS site *Bay Area Box Line 11*, which spans from northwest to southeast under clear-sky conditions. The dataset comprises orthorectified HS imagery with high spatial and spectral resolutions, facilitating environmental analysis across diverse terrain types. Base map source: Google Maps (Map data © Google, 2025).

TABLE II
QUANTITATIVE COMPARISON OF DIFFERENT MODELS ON HIG

Models	PSNR↑	ERGAS↓	SAM↓	Q Index↑	SSIM↑
MHF-Net	30.18	1.61	4.14	0.82	0.93
DBIN	31.91	1.54	3.37	0.89	0.93
Fusformer	32.14	1.57	3.05	0.88	0.94
MoG-DCN	33.34	0.80	3.18	0.93	**0.96**
MSST-Net	31.18	0.83	2.73	0.95	0.95
CASSIDiff	33.11	0.84	**2.36**	0.92	0.95
SpecSwin3D	**35.84**	**0.73**	2.39	**0.97**	**0.96**

All models were tested on 30 samples, and the results are averaged over all bands (224).

[45], and MSST-Net [35], using five standard image quality metrics: PSNR, ERGAS, SAM, Q Index, and SSIM. The results are summarized in Table II.

In terms of PSNR, SpecSwin3D achieves the highest score of *35.84 dB*, which is approximately *+2.5 dB* higher than the strongest competing feed-forward model MoG-DCN (33.34 dB) and *+5.8 dB* higher than the classical baseline MHF-Net (30.18 dB). For *ERGAS*, SpecSwin3D also obtains the lowest value (*0.73*), outperforming MoG-DCN (0.80) and MSST-Net (0.83) and substantially improving upon Fusformer (1.57), DBIN (1.54), and MHF-Net (1.61), indicating stronger global spatial–spectral consistency. Regarding *SAM*, the diffusion-based CASSIDiff attains the lowest angle (2.36°), while SpecSwin3D closely follows with 2.39° and still offers a clear margin over the other Transformer- and CNN-based baselines. For *Q Index* and *SSIM*, SpecSwin3D achieves the best or tied-best performance (*Q* Index *0.97*, SSIM *0.96*, tied with MoG-DCN), confirming that it preserves both spectral contrast and spatial structure with high fidelity. Overall, SpecSwin3D delivers the most balanced performance across all five metrics, while CASSIDiff provides competitive spectral fidelity at the cost of iterative diffusion sampling.

Fig. 8 provides a qualitative comparison for four representative HS bands spanning the RGB, NIR, SWIR-1, and SWIR-2 regions. Across all bands, SpecSwin3D reconstructs field boundaries, road edges, and fine textural patterns with sharp, spatially coherent structures, while effectively suppressing noise. CASSIDiff often produces visually competitive results but occasionally exhibits slight over-smoothing or residual speckle due to its iterative denoising process. MoG-DCN and MSST-Net generally preserve large-scale structures but tend to introduce more noise and local contrast artifacts, particularly in the NIR and SWIR bands. MHF-Net, DBIN, and Fusformer show more pronounced degradation, with noticeable blurring, loss of detail, or structured noise in high-frequency regions. Fig. 6 illustrates the spectral reconstruction performance of the proposed SpecSwin3D model for a representative vegetation pixel, covering the full HS range from the visible to the SWIR region. Compared with the baseline training strategy without cascading (Fig. 1), SpecSwin3D with cascade training maintains consistently high reconstruction accuracy even for HS bands that are spectrally distant from the MS inputs. Specifically, without cascade training, reconstruction errors increase substantially in distant spectral regions, indicating that a single-stage joint optimization struggles to capture long-range spectral dependencies.

In contrast, the proposed cascade training strategy progressively introduces harder spectral targets and effectively stabilizes learning across wavelengths, leading to smoother and more accurate spectral curves throughout the entire HS range. Although minor performance variations remain due to intrinsic spectral complexity and absorption effects, the overall degradation observed in distant bands is significantly mitigated. This comparison demonstrates that HS bands exhibit inherently

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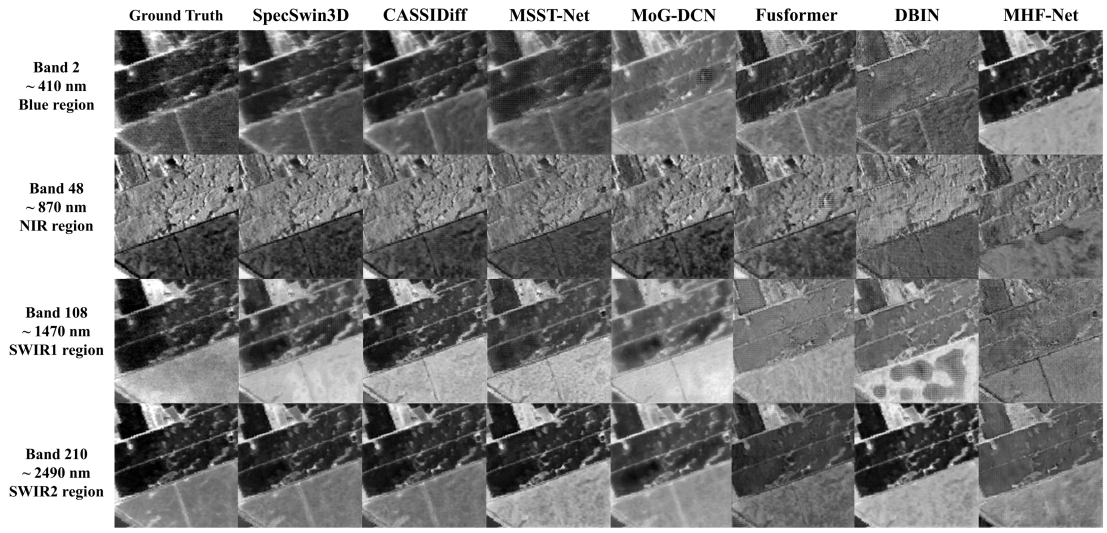


Fig. 8. Visual comparison of constructed HS bands for a selected 128×128 tile. (Top to bottom) Band 2 (410 nm, blue), Band 48 (870 nm, NIR), Band 108 (1470 nm, SWIR-1), and Band 210 (2490 nm, SWIR-2). SpecSwin3D produces the cleanest and most structurally coherent reconstructions across all spectral regions, while competing methods exhibit varying levels of noise, blurring, and texture distortion.

different reconstruction difficulties and confirms the necessity of structured cascade training and fine-tuning for achieving robust and reliable HIG across the full spectral domain.

IV. Discussion

A. Band Sequence Design

As illustrated in Fig. 3, the input depth is constrained to be a multiple of 32 due to the SpecSwin3D encoder design, which applies five successive patch merging operations along the depth dimension. Each patch merging layer halves the depth resolution, leading to a 2^5 downsampling factor and thus requiring the input depth to be divisible by 32. We further adopt the standard shifted-window attention size of $7 \times 7 \times 7$ as defined in SwinUNETR [51]. Under this setting, each Swin Transformer block attends to local 3-D neighborhoods spanning seven adjacent slices along the depth dimension, enabling effective modeling of interslice spectral correlations. We conducted a comparative analysis of two band sequence designs based on the five MS input bands [9, 20, 30, 40, 52] (see Section II-A for details). The first employs an interleaved approach [9, 20, 30, 40, 52, 20, 40, 9, 30, 52, 9, 20, 30, 40, 52, 30], ensuring that all unique band pairs co-occur within at least one adjacent slice pair along the depth dimension. The second implements a contrast repeated pattern [9, 9, 9, 20, 20, 20, 30, 30, 30, 30, 40, 40, 40, 52, 52, 52], in which all five bands are repeated three times except for band 30, yielding a 16-depth input. This design preserves equal band frequency while presenting bands in a repeated sequential order. Performance metrics across six evaluation criteria are presented in Fig. 9.

The results show that the interleaved design consistently outperforms the repeated design across nearly all bands. This superior performance suggests that the Swin Transformer mechanism significantly enhances model efficacy by effectively capturing and appropriately weighting depthwise attention within each 7×7 patch, thereby optimizing the representation of band and spectral information. Under the window-based self-attention mechanism, arranging spectral bands in tight adjacency does not reduce model expressiveness compared to looser window-only coverage. Instead, it introduces a stronger inductive bias that improves the

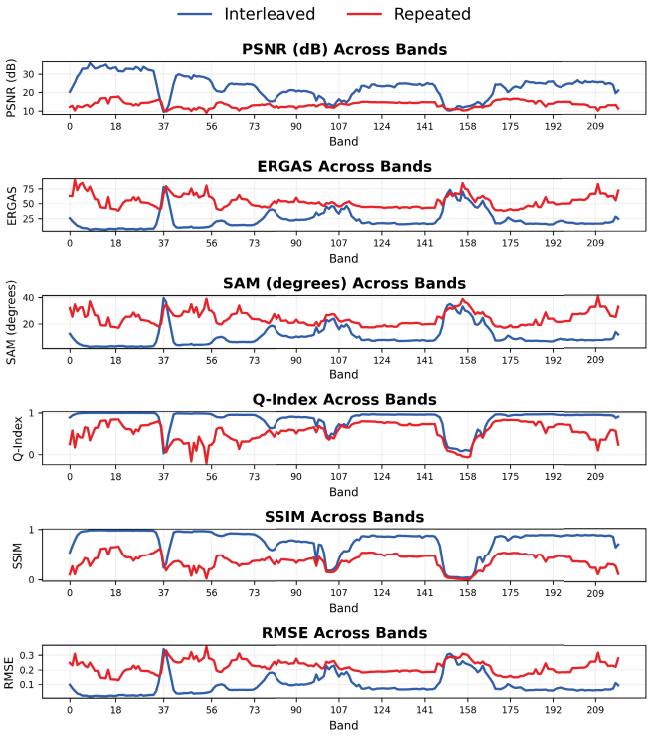


Fig. 9. Quantitative comparison of two band sequence designs—*interleaved* ([9, 20, 30, 40, 52, 20, 40, 9, 30, 52, 9, 20, 30, 40, 52, 30]) and *repeated* ([9, 9, 9, 20, 20, 20, 30, 30, 30, 30, 40, 40, 40, 52, 52, 52]), evaluated across all spectral bands using six metrics: PSNR, ERGAS, SAM, Q Index, SSIM, and RMSE. Each subplot shows the average metric values across 30 test samples. Higher values of PSNR, Q Index, and SSIM, and lower values of ERGAS, SAM, and RMSE indicate better reconstruction performance. The interleaved design consistently outperforms the repeated design across nearly all bands.

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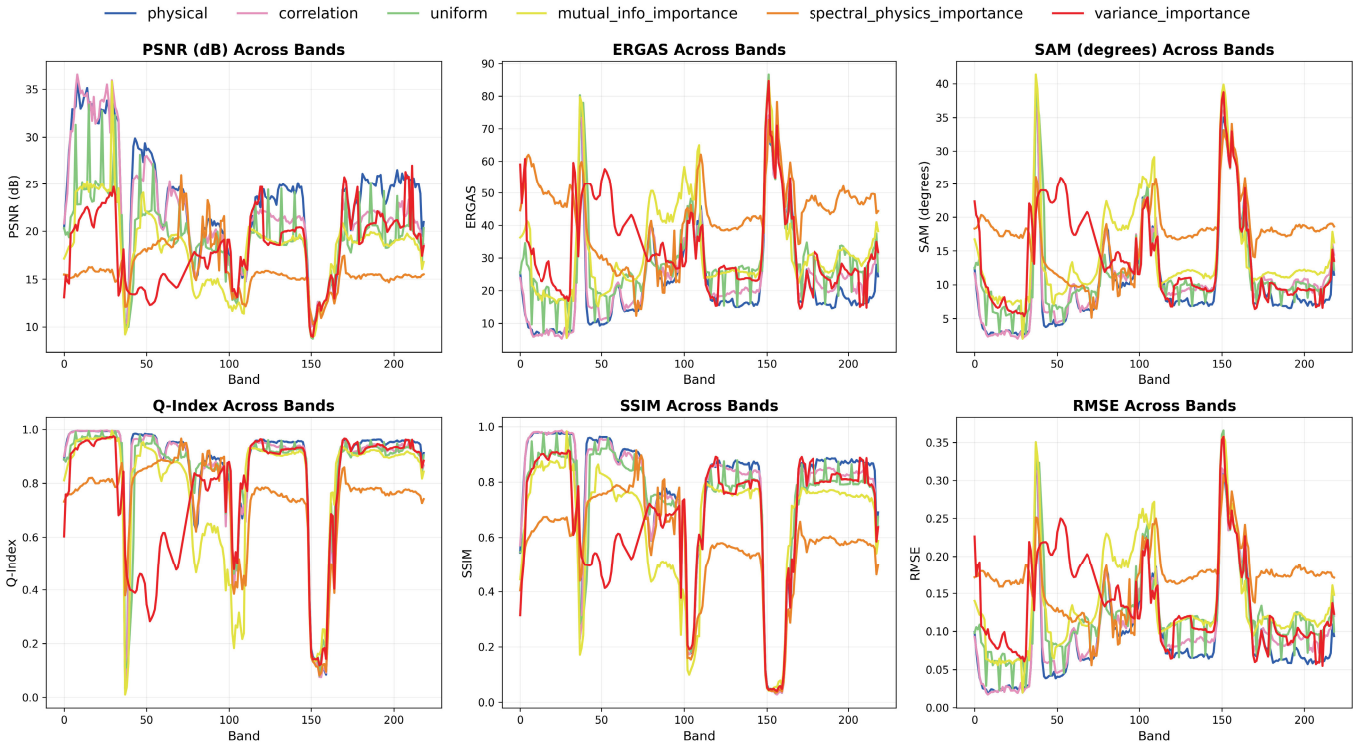


Fig. 10. Quantitative comparison of six cascade training strategies, including physical, correlation_importance, uniform, mutual_info_importance, spectral_physics_importance, and variance_importance, evaluated across all spectral bands using six metrics.

statistical efficiency of relative positional encoding and stabilizes cross-window spectral interactions. In the worst case, tight adjacency yields performance comparable to general window coverage, while, in practice, it consistently leads to improved reconstruction accuracy.

B. Curriculum-Based Cascade Training Strategies

As shown in Fig. 1, HS bands in Area II suffer from degraded reconstruction due to weak spectral correlation with the MS inputs. To address this challenge, we adopt a multistage cascade training framework that progressively expands the spectral prediction space from near MS to spectrally distant bands. A key component of this framework is the cascade curriculum, which determines the ordering and grouping of HS bands according to reconstruction difficulty and directly impacts optimization stability and knowledge transfer.

Table I summarizes six curriculum strategies, and Fig. 10 shows the comparison of their performance across the full HS range. Contrary to the intuition that prioritizing distant or uniformly distributed bands may be optimal, the physical and correlation-based curricula consistently outperform alternative strategies across nearly all bands and metrics. This result indicates that effective cascade design should emphasize physically meaningful spectral grouping and strong interband correlations, rather than spectral distance alone. By introducing harder bands that remain spectrally or statistically connected to previously learned targets, these curricula enable more reliable knowledge transfer across cascade stages.

Fig. 11 further demonstrates that cascade training consistently outperforms joint single-stage optimization across all

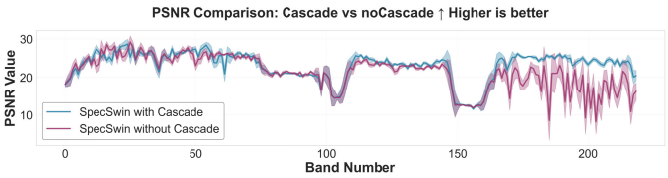


Fig. 11. PSNR comparison of cascade training versus a no-cascade baseline across all 224 HS bands, showing consistent reconstruction quality improvement particularly in distant spectral regions (Area II).

224 bands, with particularly large improvements in spectrally distant regions. These results confirm that progressively expanding the spectral prediction space facilitates more effective learning of long-range spectral relationships and significantly mitigates spectral hallucination.

C. Impact of SPE on Attention

To assess the impact of SPE on the Transformer's spectral modeling ability, we visualize the depth-to-depth self-attention matrices from the first Swin3D stage (Fig. 12), where each matrix indicates the attention a query slice (vertical axis) assigns to a key slice (horizontal axis) within the seven-depth local window. Without SPE, attention becomes diffuse and weakly structured, with adjacent and distant spectral offsets receiving similar weights. This indicates that the Transformer cannot reliably infer absolute wavelength positions from content alone, leading to spectral ambiguity—particularly for wavelengths far from the MS anchors. When SPE is introduced, the attention becomes sharply localized: the strongest

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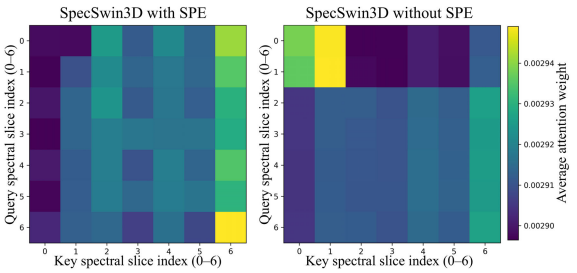


Fig. 12. Depth-to-depth self-attention comparison with and without SPEs. Each matrix shows averaged self-attention weights among the seven spectral slices within a 3-D Swin window. (Left) SpecSwin3D with SPE exhibits structured, localized attention, reflecting stronger absolute wavelength awareness and enhanced local spectral coherence. (Right) Without SPE, attention is diffuse, uniform attention distributions, indicating weaker spectral discrimination. These results demonstrate that SPE improves the Transformer's modeling of both local and long-range spectral dependencies, yielding more stable HIG.

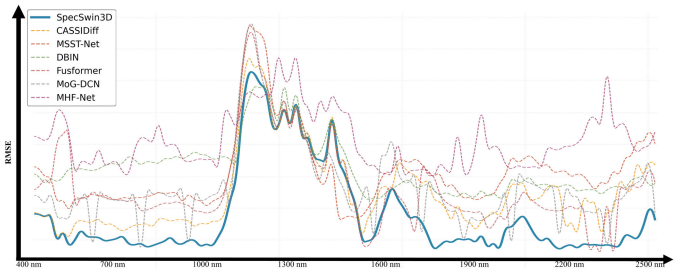


Fig. 13. Per-band RMSE comparison. SpecSwin3D maintains the lowest error and smoothest spectral behavior, while baselines exhibit strong oscillations, indicating unstable extrapolation.

TABLE III

SUMMARY OF ABLATION, GENERALIZATION, AND ROBUSTNESS RESULTS FOR SPECSWIN3D

Category	Setting	PSNR	SSIM	SAM
Ablation	SpecSwin3D (full model)	33.39	0.95	2.41
	w/o anchor consistency	33.02	0.93	2.52
	w/o PCA manifold constraint	30.78	0.92	2.50
	w/o smoothness + SAM loss	32.24	0.92	2.64
	w/o all spectral regularization	31.41	0.92	2.57
	Handcrafted ordering (w/o SPE)	27.36	0.85	3.02
	Learned SPE	34.95	0.93	2.72
Generalization	AVIRIS input, same season	34.73	0.96	2.78
	AVIRIS input, different season	35.86	0.96	2.81
	Sentinel-2 input, AVIRIS GT	32.57	0.94	2.92
Robustness	Clean input	32.66	0.94	2.46
	Gaussian noise (5%)	32.48	0.93	2.58
	Gaussian noise (10%)	30.78	0.93	2.87
	1-pixel spatial shift	32.29	0.94	2.37
	2-pixel spatial shift	31.61	0.94	2.63

responses consistently appear along the first two spectral offsets, while distant offsets receive suppressed weights. This pattern aligns with natural spectral continuity and indicates that SPE provides the global wavelength context the model cannot learn implicitly. These behavioral differences are also reflected quantitatively in Table III. Using learned SPE improves PSNR from 27.36 to 34.95 dB, boosts SSIM from 0.85 to 0.93, and reduces SAM from 3.02° to 2.72°, confirming that SPE substantially strengthens spectral discriminability.

D. Generalization and Robustness Analysis

We further evaluate the generalization capability and robustness of SpecSwin3D under cross-region, cross-season, cross-platform, and out-of-distribution (OOD) wavelength settings, as well as under controlled noise and spatial misregistration perturbations. These experiments extend beyond the primary AVIRIS-Bay area evaluation and provide additional evidence of reconstruction stability under domain shifts.

1) Cross-Season and Cross-Platform Transfer: Table III shows that SpecSwin3D generalizes well across seasonal and sensor shifts. Performance is consistent between same- and different-season AVIRIS inputs, and using Sentinel-2 MS input causes only a modest degradation (PSNR 32.57), indicating stable spectral–spatial reconstruction under domain variation.

2) OOD Wavelengths: Fig. 13 reports per-band RMSE across the full spectral range. As the wavelength distance from the input MS bands increases, reconstruction error naturally becomes higher. SpecSwin3D shows a smooth and gradual degradation trend, whereas prior methods exhibit strong oscillations and large error spikes, indicating unstable spectral extrapolation. The smooth profile of SpecSwin3D demonstrates reduced spectral hallucination and improved OOD robustness.

3) Noise and Misregistration Robustness: As shown in Table III, SpecSwin3D remains stable under moderate noise and small spatial shifts: PSNR decreases only slightly under 5% Gaussian noise and 1–2 pixel misregistration, and SSIM remains above 0.93 in all cases. These results indicate strong robustness to sensing noise and minor geometric perturbations.

E. Spectral Regularization

1) Ablation of Spectral Regularization: Table III evaluates the contribution of each spectral regularization component. Removing the anchor consistency constraint leads to a moderate reduction in PSNR and SSIM, indicating its role in maintaining alignment between reconstructed HS signatures and the MS sensor response. Eliminating the PCA manifold constraint produces a larger degradation, suggesting that restricting reconstructed spectra to a low-dimensional reflectance manifold is essential for physical plausibility. Omitting the smoothness and SAM losses further increases spectral distortion, as reflected by the higher SAM values. When all spectral regularization terms are removed, performance declines across all metrics, confirming that these components work jointly to stabilize spectral behavior and improve overall HIG fidelity.

F. Downstream Applications

Beyond HIG quality, we further assess the practical utility of SpecSwin3D for RS applications using HS or MS imagery with enhanced spatial and spectral resolutions. Spatially, SpecSwin3D leverages native HSR MS inputs to generate HSR HS bands, enhancing HS-based applications through improved spatial detail. Spectrally, SpecSwin3D enriches MS-based applications by generating intermediate HS bands, increasing

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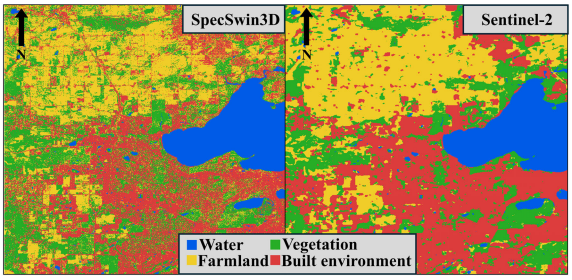


Fig. 14. Land-use segmentation results. (Left) Using SpecSwin3D (10-m bands). (Right) Using Sentinel-2 imagery (20-m bands). Both panels depict the same area in Madison, WI, USA; colors denote water (blue), vegetation (green), farmland (yellow), and built environment (red). The generated HS bands maintain comparable classification performance while substantially enhancing spatial resolution, yielding clearer boundary delineation.

TABLE IV
PERFORMANCE COMPARISON IN APPLICATIONS. RESULTS ARE REPORTED IN %

Dataset	Resolution	Accuracy	Precision	Recall	F1 Score
Hyperspectral-based: Landuse segmentation (average over classes)					
Sentinel-2	20 m	81.0	84.8	75.8	80.1
SpecSwin3D	10 m	80.6	81.5	75.1	78.1
Multispectral-based: Burnt area detection					
Sentinel-2	10 m	80.2	86.3	55.8	67.8
SpecSwin3D	10 m	86.0	88.8	85.7	87.2

spectral resolution, and enabling more informative spectral indices. Using Sentinel-2 dataset, we evaluate two downstream tasks: land-use segmentation with HS imagery and burned area detection with MS imagery. The dataset includes 10-m bands (Band 2-Blue, 490-nm center; Band 3-Green, 560 nm; Band 4-Red, 665 nm; and Band 8-NIR, 842 nm), and 20-m bands (Band 5-Red Edge 1, 705 nm; Band 6-Red Edge 2, 740 nm; Band 7-Red Edge 3, 783 nm; Band 8A-Narrow NIR, 865 nm; Band 11-SWIR1, 1610 nm; and Band 12-SWIR2, 2190 nm). *SpecSwin3D* is trained using Sentinel-2 10-m bands (2, 3, 4, 8, with band 8 duplicated) as a 3-D input sequence.

1) HS-Based Downstream Task (Land-Use Segmentation): We used the public DynamicWorld dataset [66] as our ground truth, which provides land-use labels with nine classes. We reclassified these into four categories: water, vegetation, farmland, and built environment (buildings, bare land, and roads). For classification, we employed the CatBoost algorithm [67]. The baseline model used both the 20-m Sentinel-2 bands (B5, B6, B7, B11, and B12) and the 10-m bands (B2, B3, B4, and B8), with the latter downsampled to 20 m for spatial alignment. To enhance spatial detail, we applied *SpecSwin3D*, which takes native 10-m Sentinel-2 bands (B2–B4 and B8; with B8 duplicated) as input and generates 10-m HS bands corresponding to the central wavelengths of the Sentinel-2 20-m bands (B5, B6, B7, B11, and B12). The results (see Fig. 14 and Table IV) show that SpecSwin3D-generated HS bands achieve land-use classification performance comparable to the 20-m Sentinel-2 baseline (81.0% versus 80.6% overall accuracy). Despite a very small reduction in accuracy, the SpecSwin3D model improves spatial resolution from 20 to 10 m and yields enhanced boundary delineation, particularly between farmland and built-up areas. This is particularly noticeable in northern Madison, where road networks appear significantly clearer. This finer spatial granularity offers practical advantages for downstream geospatial analysis by providing clearer and more spatially coherent land-use predictions.

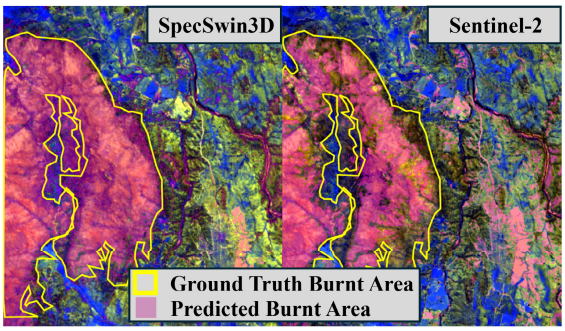


Fig. 15. Burned-area detection results. (Left) SpecSwin3D (dNBR). (Right) Sentinel-2 (dNDVI). Yellow outlines indicate the ground-truth burned perimeter; magenta shading shows the predicted burned area. Both panels depict the same location. Spectral enrichment via SpecSwin3D-generated SWIR2 enables NBR computation, dramatically improving burned area detection accuracy and recall over NDVI-only approaches.

2) MS-Based Downstream Task (Burned Area Detection): We detect burned areas using the BiAU-Net dataset [68], a bitemporal Sentinel-2 dataset containing 10-m bands (B2–B4 and B8) before and after wildfire events. A threshold-based NDVI difference measure ($\text{NDVI}_{\text{pre}} - \text{NDVI}_{\text{post}} > \text{threshold}$) is used as the baseline detector. To improve sensitivity, we apply *SpecSwin3D* to generate HS bands in the SWIR2 region at 10-m resolution, enabling computation of the NBR difference ($\text{NBR}_{\text{pre}} - \text{NBR}_{\text{post}} > \text{threshold}$) [69], [70]. As shown in Fig. 15 and Table IV, the NDVI-based baseline struggles with limited recall (55.8% at 80.2% accuracy). In contrast, incorporating SpecSwin3D-generated SWIR2 bands (10 m) enables NBR computation and substantially boosts detection performance to 86.0% accuracy and 85.7% recall. This demonstrates the value of spectral enrichment: by providing access to wavelengths absent from native Sentinel-2 inputs, even simple threshold-based methods achieve markedly improved burned-area detection.

V. CONCLUSION AND FUTURE WORK

This article presents *SpecSwin3D*, a Transformer-based framework for HIG that explicitly models physical, spatial, and spectral characteristics of surface reflectance. By integrating a U-Net architecture with a 3-D shifted-window Transformer encoder, SpecSwin3D jointly captures spatial structures and spectral dependencies, achieving consistent performance improvements over state-of-the-art HIG methods across the full HS range.

Sequence Optimization and Spectral Representation: This work demonstrates that band sequence design is a fundamental representation choice for HIG. By organizing MS inputs along the spectral depth dimension, SpecSwin3D enables 3-D self-attention to effectively capture structured local and long-range interband relationships, providing an inductive bias that

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promotes spectral continuity and physical consistency. Compared with channel-based formulations that collapse spectral structure into 2-D feature maps, the proposed depth-oriented representation preserves spectral ordering throughout the encoder and facilitates more stable spectral–spatial learning. Future research will explore more flexible spectral representations beyond fixed band sequences, including hybrid designs that combine depth-based ordering with channelwise feature fusion, as well as adaptive spectral encodings that better accommodate diverse sensor characteristics and scene variability.

Cascade Curriculum Design: The proposed cascade curriculum highlights the importance of structured optimization for HIG under extremely sparse MS inputs. By progressively expanding reconstruction targets from spectrally proximal bands to more distant regions, cascade training effectively mitigates spectral hallucination and stabilizes learning across the wavelength domain. The experimental results confirm that HS bands exhibit heterogeneous reconstruction difficulty and that curriculum-based training yields substantial gains, particularly for spectrally distant bands. In this study, cascade levels are constructed using deterministic physical priors and data-driven statistics, offering an interpretable and efficient training strategy. Future work will investigate adaptive or self-optimizing curricula that dynamically adjust band ordering and pacing across scenes and land-cover types, further improving robustness and generalization.

Overall, SpecSwin3D establishes a unified framework that integrates representation-aware spectral modeling with curriculum-based optimization, providing a scalable and physically grounded solution for HIG. We expect the proposed principles to generalize to broader problems in spectral super-resolution and cross-modal remote sensing analysis.

Data and Code Availability

The datasets and source code for SpecSwin3D are publicly available at https://github.com/scdmlab/SpecSwin3D

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